

Integrated Approach for the Identification of KRAS Inhibitors, Utilizing Machine Learning

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Abstract—The aim of this study is the discovery of potential KRAS inhibitors by evaluating the bioactivity of ligands derived from virtual screening through a machine learning model. The training dataset consisted of 2290 compounds that were tested in vitro and characterized as either “active” or “non-active”. The model was evaluated internally using k-fold cross validation (k=10) and externally via the hold-out method. Out of 6 classifiers tested, “AdaBoost” was the one that generated the highest accuracy (71%) and was subsequently employed for the classification. The accuracy across all classifiers was relatively low, ranging from 65% to 70%, which is attributed to the specific dataset used in the training process. Pharmit results were based on 5 pharmacophore features, along with parameters related to the number of rotational bonds and the molecular weight. From the 1391 ligands identified by pharmit, 500 were classified as “active” and 891 as “non-active”. The RMSD of the top 10 ligands ranged from 0.357 to 0.473. The integrated approach followed in this study ensures better results than virtual screening alone, because it takes into consideration not only pharmacophore features but also in vitro observations translated into molecular descriptors.

Keywords— AdaBoost, Bioactivity, Classification, Descriptors, KRAS, Machine Learning Model, Mutations, Pharmacophore Features, RMSD, Virtual Screening

I. INTRODUCTION

RAS is a proto-oncogene that plays a crucial role in cell proliferation. It was discovered nearly 45 years ago and its mutations are among the most repeatedly observed in cancer cases related to oncogenes, with them being linked with more than 20% different types of cancer [1], [2]. There are 3 members in the RAS family, NRAS, HRAS, and KRAS, with the last one being the most frequently mutated (81%), followed by NRAS (15%) and HRAS (4%) [2], [3]. KRAS mutations have been observed in more than 95% of pancreatic cancer cases, 55% of thyroid cases and 35% of lung and colorectal cancer cases [4].

KRAS is an enzyme located on the inner side of the plasmatic membrane. Its main function is the transmission of signals related to cell growth and differentiation from the cell surface to the cytoplasm. KRAS is a guanosine triphosphate hydrolase (GTPase) with two possible states, the active state, in which the protein is bound with GTP and the inactive state, in which it is bound with GDP [5]. Depending the state, the protein structure and more specifically the areas named Switch I (residues 30-40) and Switch II (residues 60-76) change allowing the binding with other compounds [2], [6].

The transition from the active to the inactive state is becoming through the GTP to GDP hydrolysis. This requires the interaction between KRAS and GTPase Activating Proteins (GAP), which accelerate the hydrolysis. In the active state, RAF binds to KRAS, which leads to the temporarily activation of pathways related to cell proliferation, such as mitogen-activated protein kinase (MAPK) and phosphatidylinositol 3-kinase-protein kinase B (PI3K) signaling pathways [5], [7], [8].

In case of mutations, the KRAS structure changes, preventing the interaction between GAP and KRAS itself and consequently

the hydrolysis. As a result, the enzyme stays continuously in the active state, activating the pathways related to cell growth.

The most regularly mutated gene location is the codon G12, with G12C and G12D being two of the most predominantly sequence alterations in RAS gene [2]. In the first one, the glycine (G) residue in the 12th position is replaced by cysteine (C). G12C is not as common compared to G12D but it has very poor survival rates compared to other KRAS mutations [9]. The G12D mutation is the most frequently observed with approximately 34% occurrence in the overall pancreatic cancer cases. In this mutation, the glycine (G) residue in the 12th position is replaced by aspartate (D) [10].

In the past, there were several attempts to identify KRAS inhibitors, but the majority of them resulted in a stalemate. The development of inhibitors that mimic the ATP-competitive inhibitors in kinases was an early concept that soon proved to be unfeasible [2]. Kinases bind with ATP with low affinity, making it possible for other compounds to substitute ATP and alter kinase activity, whereas GTP binds with much higher affinity with KRAS, hence preventing the inhibition at the orthosteric center [2], [11]. Another idea was the finding of compounds that could substitute the function of GAPs in hydrolysis. At first, the results were not promising, but recently the idea has been reevaluated due to the observation that RGS proteins can interact with the mutant KRAS [12].

Due to the above, scientists soon lost interest in finding direct KRAS inhibitors and characterized the KRAS as “undruggable” [2]. However, the identification of a binding pocket in 2012, rekindled hope, making the allosteric inhibition possible [6], [13].

This study aims at the identification of ligands that can bind to this pocket, which is located near Switch II, and alter the mutant KRAS structure. Hence, preventing the interaction between RAF and KRAS and the subsequent activation of the pathways.

II. MATERIALS AND METHODS

The study is divided into two distinct phases, the first one is the virtual screening of compounds against KRAS, and the second one comprises the building of a machine learning model that is used as an additional filter for the virtual screening results.

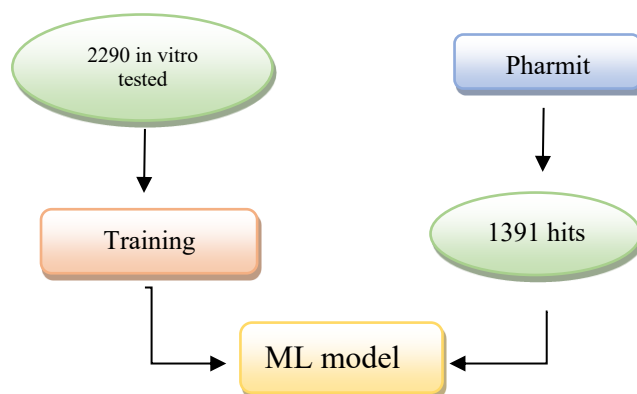


Fig.1 Schematic representation of the pipeline.

A. Virtual Screening

In this phase, a search for possible inhibitors was conducted by employing Pharmit search engine. The crystalized structure used was retrieved from the protein databank (PDB) with code “4EPY”, and the selected database was “MolPort”.

The identification is based on parameters such as pharmacophore features, the maximum molecular weight, and the maximum number of rotation bonds. The selected pharmacophore features used were two hydrogen donors with coordinates (23.7, -10.4, 14.2) and (24.4, -6.8, 14.8), a hydrogen acceptor with coordinates (23.3, -1.3, 7.1), a hydrophobic interaction with coordinates (22.8, 0.9, 9.0), and an aromatic interaction with coordinates (25.1, -4.2, 13.5). For better results, further parameters were defined including a maximum molecular weight of 550 and a maximum of 8 rotational bonds.

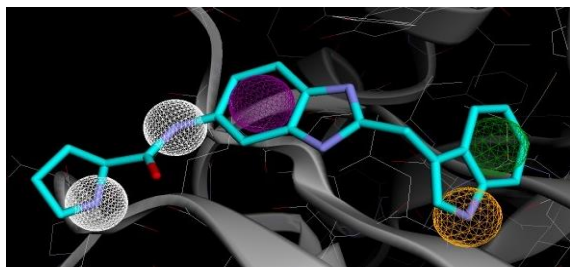


Fig. 2 The selected pharmacophore features in “pharmit”.

The hits resulted from Pharmit were a list of compounds in ascending RMSD order, that comprised the test set used in the machine learning model.

B. Machine Learning Model

The building of the machine learning model is divided into 4 stages, data collection, data pre-processing, model training, and model evaluation.

The dataset used in the training process came from “ChEMBL”, a bioactivity database used in drug discovery, and it was the “ChEMBL 3736”. The selected dataset included information for about 2290 compounds that were tested in vitro and it consisted of the molecule ChEMBL ID, molecule’s Simplified Molecular Input Line Entry System (SMILES), a text-based representation of a chemical structure, and lastly a characterization as active or non-active depending on the in vitro experiments.

Data pre-processing was achieved using the “Spyder” Integrated Development Environment (IDE) with python as the programming language. Part of the process was the extraction of descriptors (features) from the SMILES using the RDKit tool, the standardization of the different descriptors values and the application of the SMOTE method for achieving unbiased results. The SMOTE method is employed when the data in one of the classes are considerably fewer than those in the other class.

In the training stage, six classifiers were tested and the one with the higher classification ability was selected. The classifiers employed were the “RandomForest”, the “CART”, the “XGBoost”, the “ExtraTrees”, the “AdaBoost”, and the “GradientBoost”. The maximum number of descriptors used in the training was set at 15, and the process was repeated for 5 epochs to ensure statistical validity.

The model was evaluated both internally, with the 10-fold cross-validation method, and externally, with the hold-out method.

C. Classification of Pharmit Results

The last step included the conversion of the SDF file with the hits that came from Pharmit to an Excel file, and the extraction

of descriptors using the RDKit tool as well. Then, the pharmit results were fed into the machine learning model to be classified as “active” or “non-active” ligands.

III. RESULTS

Pharmit search resulted in 1391 possible inhibitors with RMSD in range of 0.357 to 0.945.

The extracted descriptors from the training dataset and the CSV file were 217 in total, of which 46 had high frequency, and thus high importance. The high-frequency features are defined as the descriptors that ranked in the top 20 more times than the number of epochs. In this study, 5 epochs were defined, meaning that 46 descriptors ranked in top 20 more than 5 times out of the overall 30 runs (number of epochs multiplied with the number of classifiers tested). From these, 15 descriptors were set as decisive factors for the classification of the compounds as active or non-active.

TABLE I TOP 5 DESCRIPTORS WITH HIGH IMPORTANCE

	Descriptor	Frequency	Explanation
1	MolLogP	72	Measure of the overall lipophilicity
2	BCUT2D LOGPHI	72	Topological measure of the lipophilicity distribution
3	PEOE_VSA7	58	Measure of the surface area of nearly neutral atoms
4	BCUT2D MWLOW	58	Topological measure of the distribution of atomic weights
5	AvgIpc	57	Topological measure of the structural complexity

Table. 1 Top 5 description that participated in the classification process.

“AdaBoost” demonstrated the highest accuracy (71%) among the 6 classifiers, and it was the one that was selected to classify Pharmit hits as active or non-active ligands.

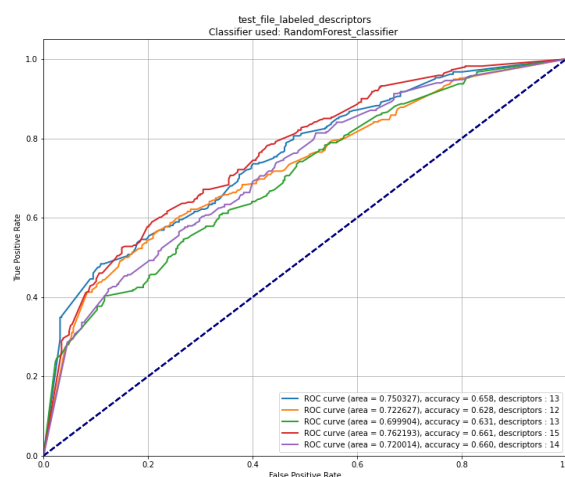


Fig.3 ROC curve diagram and accuracy of the “RandomForest” classifier.

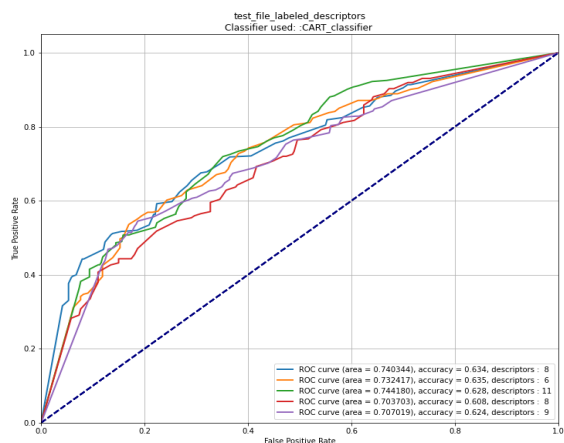


Fig.4 ROC curve diagram and accuracy of the “CART” classifier.

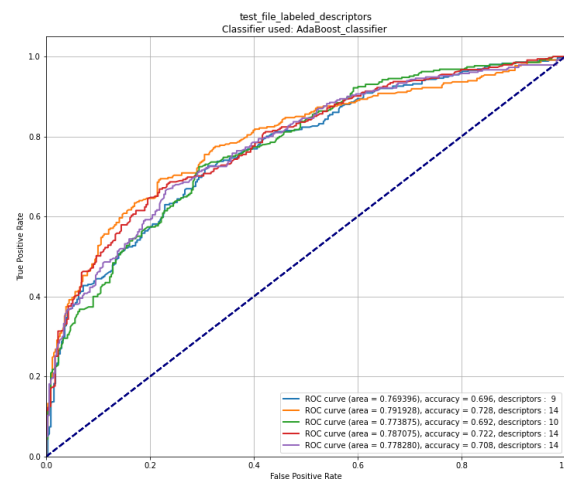


Fig.7 ROC curve diagram and accuracy of the “AdaBoost” classifier.

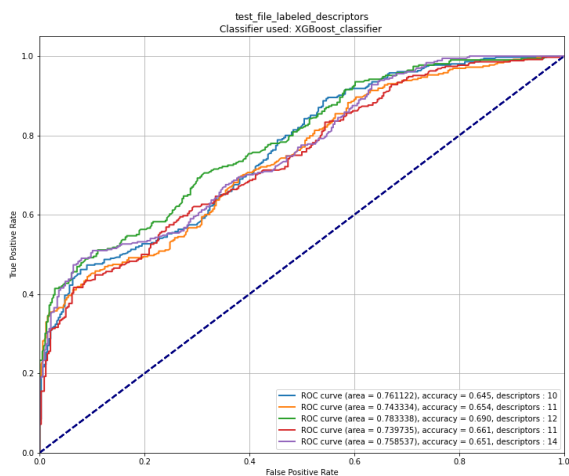


Fig.5 ROC curve diagram and accuracy of the “XGBoost” classifier.

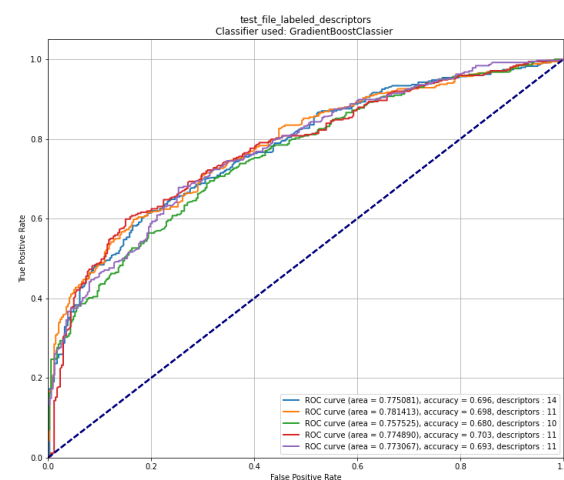


Fig.8 ROC curve diagram and accuracy of the “GradientBoost” classifier.

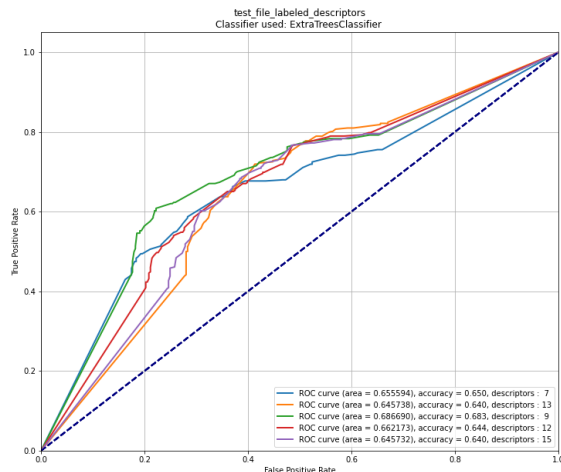


Fig.6 ROC curve diagram and accuracy of the “ExtraTrees” classifier.

The machine learning model classified 500 compounds as “active” and 891 compounds as “non-active”. Table II lists the top 10 ligands that are likely to have an inhibitory effect against mutant KRAS.

TABLE II TOP 10 LIGANDS

Overall Rank	Pharmit Rank	Name	RMSD
1	1	MolPort-039-347-422	0.357
2	2	Molport-051-896-091	0.367
3	3	MolPort-046-620-676	0.378
4	7	Molport-051-820-872	0.397
5	8	MolPort-019-894-132	0.412
6	14	MolPort-000-943-528	0.455
7	15	MolPort-045-962-970	0.461
8	16	MolPort-042-568-081	0.465
9	19	MolPort-020-273-826	0.471
10	20	MolPort-030-049-475	0.473

Table. 2 Top 10 ligands from the machine learning model and the virtual screening.

IV. DISCUSSION

It is observed from the graphs (Fig. 3-8) that a large number of features is not always optimal and does not necessarily translate into higher accuracy in the classification process.

The results derived from the machine learning model are not the expected. The calculated accuracy across all six classifiers is relatively low ($\approx 65\%$ - 70%). This is attributed to the heterogeneous origin of the samples in the dataset used in the training process, and the fact that the characterization “active” refers to ligands that can bind and interact with various mutant forms of KRAS.

Also, the Pharmit top 10 hits were not consistent with the top machine learning results. This is because, Pharmit determines the bioactivity ranking based on a limited number of pharmacophore characteristics and parameters. In addition, in virtual screening it is assumed that the enzyme is still in a specific crystallized pose and not a dynamic molecule, making it quite likely for the Pharmit results to differ from in vitro observations depending on ligand bioactivity. These are the main reasons why Pharmit results alone might be misleading regarding the discovery of new drugs.

V. CONCLUSIONS

In this study, virtual screening is integrated with machine learning methodologies to ensure improved identification of KRAS inhibitors. The machine learning model is an additional filter that can predict compounds’ bioactivity based on in vitro experiments. The selected classifier was “AdaBoost” with an accuracy of 71%. This relatively low value is likely attributed to the heterogeneous origin of the samples in the training set. Overall, 500 ligands were classified as active and 891 were classified as non-active.

VI. REFERENCES

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